

LANSING, IL, USA

WMO: 722126

Lat: 41.540N

Lon: 87.532W

Elev: 616

StdP: 14.37

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04879

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.9	(c) 4.3	(d) -9.7	(e) 3.3	(f) 0.8	(g) -5.3	(h) 4.2	(i) 5.2	(j) 27.6	(k) 30.7	(l) 24.7	(m) 33.7	(n) 7.6	(o) 270	(p) 0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	90.6	74.2	88.3	73.1	85.6	71.6	77.3	86.4	75.5	83.7	74.2	82.2	10.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 74.6	(b) 132.7	(c) 82.1	(d) 73.0	(e) 125.1	(f) 79.9	(g) 71.9	(h) 120.5	(i) 79.2	(j) 41.0	(k) 86.1	(l) 39.4	(m) 83.5	(n) 38.1	(o) 81.9	(p) 83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	20.5	18.3	DB	-9.1	95.1	7.4	3.3	-14.4	97.5	-18.7	99.4	-22.9	101.3	-28.3	103.8
			WB	-9.7	80.3	7.1	2.0	-14.8	81.7	-18.9	82.9	-22.9	84.1	-28.1	85.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	25.0	27.2	38.7	49.4	59.7	69.7	74.0	71.8	66.0	53.5	40.9	30.4
	DBStd	19.38	11.64	11.23	10.86	9.03	9.23	6.83	5.62	5.07	7.25	9.05	9.49	10.41
	HDD50	2907	775	640	383	118	16	0	0	0	0	65	298	612
	HDD65	6094	1239	1058	818	472	221	31	3	7	73	376	722	1074
	CDD50	3148	1	2	34	102	317	591	743	674	481	172	27	4
	CDD65	857	0	0	3	5	57	172	281	217	104	18	0	0
	CDH74	7399	0	0	27	72	554	1526	2547	1697	833	143	0	0
	CDH80	2389	0	0	4	7	168	510	932	484	254	30	0	0
(14) Wind	WSAvg	7.6	9.4	8.7	8.8	9.2	7.4	6.5	5.7	5.1	5.7	7.6	8.7	8.8
(15) Precipitation	PrecAvg	35.4	2.1	1.8	2.2	3.3	4.1	3.9	3.9	3.9	2.8	3.0	2.6	1.9
	PrecMax	48.1	5.1	5.9	4.9	6.9	7.5	10.9	9.7	8.8	9.4	8.5	6.7	4.3
	PrecMin	20.5	0.6	0.5	0.6	1.3	0.7	0.9	1.2	0.9	0.7	0.9	0.4	0.3
	PrecStd	6.3	1.2	1.1	1.1	1.4	1.7	2.0	1.9	2.0	1.9	1.9	1.6	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.9	61.2	78.5	81.2	90.0	92.5	95.9	91.0	91.0	84.0	70.1	60.6
		MCWB	56.2	53.8	63.9	61.4	70.7	73.7	76.3	74.4	73.3	70.0	58.3	57.7
	2%	DB	50.3	53.7	68.1	75.2	84.4	89.5	90.9	88.3	87.1	78.3	63.7	54.6
		MCWB	47.3	49.4	58.1	60.4	68.7	72.8	75.5	73.3	72.3	65.6	55.8	51.3
	5%	DB	45.3	46.3	62.1	71.7	80.7	85.9	88.3	85.1	82.3	72.9	59.8	48.9
		MCWB	42.8	42.9	54.8	58.8	66.1	71.1	74.1	71.6	69.2	62.7	53.5	46.3
	10%	DB	40.0	41.1	55.3	66.0	75.5	82.2	85.1	82.3	79.0	68.5	54.9	44.7
		MCWB	37.6	37.7	49.0	55.2	64.4	69.3	72.2	70.5	67.5	61.3	49.4	41.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.7	56.4	64.8	66.0	74.3	78.4	80.4	78.2	76.6	72.4	61.6	58.0
		MCDB	58.3	58.4	75.4	75.1	85.3	88.3	89.9	86.8	86.0	81.3	65.4	59.9
	2%	WB	48.2	49.9	60.2	62.5	71.0	75.3	78.0	75.9	74.4	68.7	58.0	52.0
		MCDB	49.4	53.3	66.2	71.5	80.6	84.0	87.5	83.4	82.8	74.5	63.2	53.8
	5%	WB	42.6	43.5	55.6	59.9	68.5	73.5	76.2	74.5	72.1	65.5	53.8	46.2
		MCDB	44.9	46.2	60.9	68.6	76.9	82.8	84.8	81.4	79.2	69.3	58.5	48.2
	10%	WB	38.0	38.1	49.7	56.6	66.3	71.7	74.5	72.9	69.9	62.0	50.4	41.7
		MCDB	39.9	40.6	55.6	66.1	73.8	80.2	82.6	79.6	76.4	67.5	54.8	44.2
(35) Mean Daily Temperature Range	MDBR		14.2	15.0	16.2	19.3	19.6	19.3	18.5	18.2	19.3	18.0	15.4	12.9
	5% DB	MCDBR	18.1	21.0	24.6	27.5	24.6	22.2	21.5	21.4	22.8	23.9	21.2	17.3
		MCWBR	16.1	17.5	16.7	16.0	12.4	10.2	9.2	9.5	10.2	13.5	14.9	15.4
	5% WB	MCDBR	17.0	19.8	22.5	24.7	21.5	19.5	19.0	17.0	19.2	19.3	18.8	16.4
(40) Clear-Sky Solar Irradiance	taub		0.302	0.327	0.352	0.408	0.438	0.443	0.457	0.452	0.423	0.363	0.322	0.302
	taud		2.430	2.350	2.367	2.228	2.198	2.232	2.191	2.199	2.247	2.417	2.485	2.470
	Ebn at Noon		270	278	284	273	267	265	260	257	256	262	260	260
	Edh at Noon		25	31	35	43	45	44	45	44	39	29	23	22
(44) All-Sky Solar Radiation	RadAvg		523	794	1141	1466	1668	1920	1931	1680	1398	925	591	436
	RadStd		55	106	83	120	145	123	94	96	117	96	62	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page